



ASSIGNMENT No: 5

TITLE: Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Problem Statement

Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Learning Objectives

To implement dynamic data storage and string manipulation using ArrayList, Vector, and StringBuffer.

Theory

ArrayList and Vector are dynamic array implementations provided by the Java Collections Framework. ArrayList offers faster performance in single-threaded applications, whereas Vector provides synchronized operations for thread safety. StringBuffer is a mutable sequence of characters that allows efficient string modification without creating new objects. These classes are commonly used in applications requiring dynamic memory allocation and efficient string processing.

Code

```
import java.util.ArrayList;
import java.util.Scanner;

public class TodoList {

    public static void main (String a[]){

        ArrayList<String> courses = new ArrayList<>();
        Scanner sc = new Scanner(System.in);

        System.out.println("Enter number of tasks:");
        int n =sc.nextInt();
        sc.nextLine();

        for (int i=0; i < n;i++){

            System.out.println("Enter task:"+(i+1)+":");
            courses.add(sc.nextLine());
```

```

    }

    StringBuffer sb = new StringBuffer();
    sb.append("To-Do-List :\n");

    for (int i=0; i < courses.size();i++){

        sb.append(i + 1).append(" ").append(courses.get(i)).append("\n");

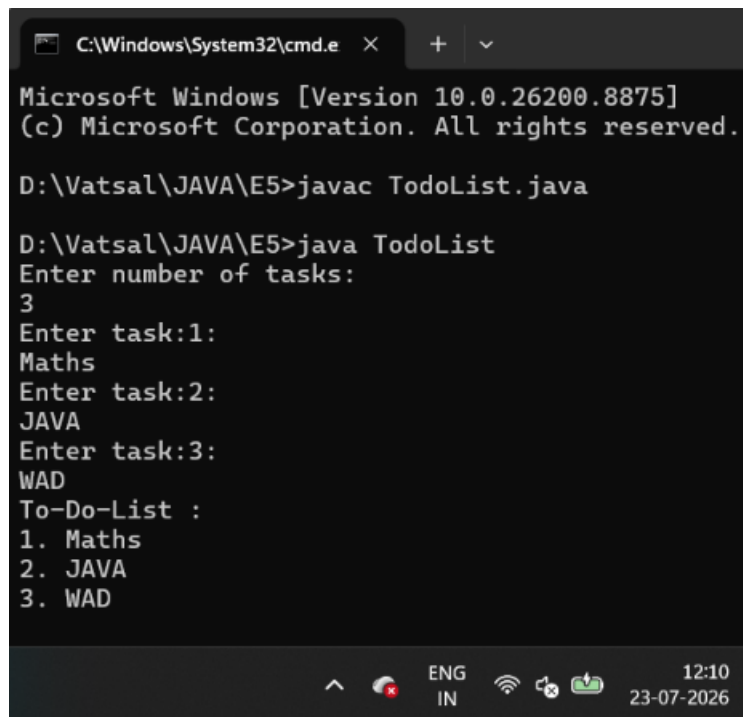
    }
    System.out.println(sb);
    sc.close();

}

}

```

Output



```

C:\Windows\System32\cmd.e  X  +  v
Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

D:\Vatsal\JAVA\E5>javac TodoList.java

D:\Vatsal\JAVA\E5>java TodoList
Enter number of tasks:
3
Enter task:1:
Maths
Enter task:2:
JAVA
Enter task:3:
WAD
To-Do-List :
1. Maths
2. JAVA
3. WAD

```

Exercise

1. Create a **To-Do List application** using ArrayList to store tasks and StringBuffer to display tasks.

Code

```
import java.util.ArrayList;
import java.util.Scanner;

public class ToDoListE {

    public static void main(String[] args) {

        ArrayList<String> tasks = new ArrayList<>();
        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter the number of tasks to add: ");
        int taskCount = scanner.nextInt();
        scanner.nextLine();

        for (int i = 0; i < taskCount; i++) {
            System.out.print("Enter task " + (i + 1) + ": ");
            String task = scanner.nextLine();
            tasks.add(task);
        }

        StringBuffer sb = new StringBuffer();
        sb.append("TO-DO LIST\n");

        if (tasks.isEmpty()) {
            sb.append("No tasks in the list.\n");
        } else {
            for (int i = 0; i < tasks.size(); i++) {
                sb.append(i + 1)
                    .append(". ")
                    .append(tasks.get(i))
                    .append("\n");
            }
        }

        System.out.println(sb.toString());

        scanner.close();
    }
}
```

Output

```
D:\Vatsal\JAVA\E5>javac ToDoListE.java

D:\Vatsal\JAVA\E5>java ToDoListE
Enter the number of tasks to add: 5
Enter task 1: JAVA
Enter task 2: WAD
Enter task 3: Maths
Enter task 4: DSA
Enter task 5: FCC
TO-DO LIST
1. JAVA
2. WAD
3. Maths
4. DSA
5. FCC

D:\Vatsal\JAVA\E5>
```

2. Create a **Student Course Registration System** using **ArrayList** to store the list of courses registered by a student and **StringBuffer** to generate and display the registered course list. The application should allow users to add, remove, and view registered courses.

Code

```
import java.util.ArrayList;
import java.util.Scanner;

public class StudentCourseRegistration {

    public static void main(String[] args) {
        ArrayList<String> courses = new ArrayList<>();
        Scanner sc = new Scanner(System.in);
        int choice;

        do {
            System.out.println("\n1. Add Course");
            System.out.println("2. Remove Course");
            System.out.println("3. Display Courses");
            System.out.println("4. Exit");
            System.out.print("Enter choice: ");
            choice = sc.nextInt();
            sc.nextLine();
        }
```

```

switch (choice) {
    case 1:
        System.out.print("Enter course name: ");
        courses.add(sc.nextLine());
        System.out.println("Course added successfully.");
        break;

    case 2:
        System.out.print("Enter course name to remove: ");
        String courseToRemove = sc.nextLine();
        if (courses.remove(courseToRemove)) {
            System.out.println("Course removed successfully.");
        } else {
            System.out.println("Course not found.");
        }
        break;

    case 3:
        StringBuffer sb = new StringBuffer();
        sb.append("\nREGISTERED COURSES\n");
        for (int i = 0; i < courses.size(); i++) {
            sb.append(i + 1).append(" ").append(courses.get(i)).append("\n");
        }
        System.out.println(sb.toString());
        break;

    case 4:
        System.out.println("Exiting...");
        break;

    default:
        System.out.println("Invalid choice!");
}
} while (choice != 4);

sc.close();
}
}

```

Output

```
C:\Windows\System32\cmd.e x + v

D:\Vatsal\JAVA\E5>javac StudentCourseRegistration.java

D:\Vatsal\JAVA\E5>java StudentCourseRegistration

1. Add Course
2. Remove Course
3. Display Courses
4. Exit
Enter choice: 1
Enter course name: JAVA
Course added successfully.

1. Add Course
2. Remove Course
3. Display Courses
4. Exit
Enter choice: 1
Enter course name: WAD
Course added successfully.

1. Add Course
2. Remove Course
3. Display Courses
4. Exit
Enter choice: 3

--- REGISTERED COURSES ---
1. JAVA
2. WAD

1. Add Course

19:39
23-07-2026

C:\Windows\System32\cmd.e x + v

1. Add Course
2. Remove Course
3. Display Courses
4. Exit
Enter choice: 2
Enter course name to remove: WAD
Course removed successfully.

1. Add Course
2. Remove Course
3. Display Courses
4. Exit
Enter choice: 3

--- REGISTERED COURSES ---
1. JAVA

1. Add Course
2. Remove Course
3. Display Courses
4. Exit
Enter choice: 4
Exiting...
```